

Health in a world of extreme heat

The heatwaves and wildfires that swept across southern Europe and the USA this summer have been traumatic and deadly. 600 excess deaths were reported in the Pacific Northwest in the USA alone. According to the Sixth Assessment Report of Working Group I of the Intergovernmental Panel on Climate Change, which synthesises the latest scientific evidence on climate change, global temperature is expected to reach or exceed 1.5°C of warming, averaged over the next 20 years. The report predicts that these record-breaking temperatures will be more frequent, intense, and last longer because of human-induced climate change. What the health consequences of a hotter climate are and how extreme heat is managed will be two of the defining questions of this decade, and are addressed by several papers in this issue of *The Lancet*.

A two-part modelling study by Katrin Burkart and colleagues from the Global Burden of Diseases, Injuries, and Risk Factors Study is the first comprehensive analysis of the global cause-specific temperature-attributable burden of deaths. Analyses of data from 65 million deaths and temperature estimates in nine countries indicate that extreme heat and cold are associated with 17 causes of death—largely cardiorespiratory or metabolic disease, but also suicide and several types of injury. The authors estimate that 1.7 million deaths worldwide in 2019 were linked to extreme heat and cold (356 000 were related to heat). Although these global estimates carry inherent uncertainties, the study shows the importance of temperature as a global health risk factor, and provides valuable data to assist policy makers and others in planning interventions. The latest evidence on how exactly heat directly affects the human body (causing ill health through dehydration or cardiovascular strain) and heat-vulnerable settings is presented in *The Lancet's* Heat and Health Series, co-led by Kristie Ebi, Ollie Jay, and Anthony Capon. The Series also reviews effective prevention and response measures to heat extremes. Two major lessons can be distilled.

First, more sustainable and affordable cooling interventions are needed that do not further increase greenhouse gas emissions or exacerbate existing inequalities. For example, despite the immediate health benefits of air conditioning and its increasing prevalence

worldwide, it is an unsustainable solution to extreme heat, contributing to air pollution, greenhouse gas emissions, and an increased urban heat island effect. Moreover, air conditioning is often unaffordable for those people most at risk from extreme heat, which widens inequality and punishes those who often contribute the least to climate change. Evidence-based strategies that focus on cooling the individual, such as use of battery-driven fans, not solely on cooling the surrounding air, must be prioritised.

Second, putting health at the centre of multisectoral policy and planning can deliver benefits beyond simply reducing heat-related burden. Because extreme heat has greater effects on the health of people in urban environments, green spaces are crucial for cooling in cities, but also provide co-benefits: they reduce exposure to air and noise pollution, relieve stress, provide a setting for social interaction and physical activity, and sequester carbon.

To reduce the impacts of extreme heat on individuals and vulnerable populations, some governments have adopted national or local heat health action plans. However, the plans are highly variable, and implementation is uneven. Many health-care facilities are unprepared. If their physical infrastructure fails because of overheating, the ability to treat patients affected by extreme heat could be compromised when needed most. All countries, local communities, and institutions need to adopt effective heat health action plans tailored to local conditions.

Such action plans provide measures to adapt to extreme heat, but do not tackle a root cause of heat extremes—greenhouse gas emissions. Minimising the effects of temperature on health requires both prevention of and adaptation to a changing climate—two of the key aims of COP26, taking place in Glasgow, UK, on Oct 31–Nov 12. But there is a danger that promises and pledges to tackle the climate crisis will not be reflected in decisive action and effective policies. Leaders at the meeting must provide transformative regulatory frameworks, policy, and investment to ensure sustainable climate change mitigation and adaptation. The papers we publish today provide a strong scientific argument that the health dimensions of heat can no longer be overlooked. ■ *The Lancet*



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For the **Global Burden of Disease Study on extreme temperature** see [Articles](#) page 685

For the **Heat and Health Series** see [Series](#) pages 698 and 709

For more on the **Pacific Northwest heatwave** see <https://www.nytimes.com/interactive/2021/08/11/climate/deaths-pacific-northwest-heat-wave.html>

For the **Sixth Assessment Report of Working Group I of the Intergovernmental Panel on Climate Change** see <https://www.ipcc.ch/report/ar6/wg1/>
For more on **COP26** see <https://ukcop26.org/>